

Addendum — Inherited-Weight Chaining

Variant of the $b_1 \leftrightarrow b_2$ chaining scheme with propagation of the weight $N_2 \rightarrow N_1$ from one step to the next

Companion to the report "Diophantine structure of the pool $b(t,k,\sigma)$ "

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0. Preliminary correction — anchor counts

A re-reading of the supplied files (chainage_beta03.txt, chainage_beta05.txt) together with an independent recomputation of the pool reveal that the table in Sections 7.1/7.2 of the previous report contained incorrect anchor counts (off by a factor of 10, too low). The correct values, verified here by two independent methods (re-reading the files and regenerating the pool), are:

Interval	Anchors reported (previous report)	Anchors actual (recomputed and verified)
$\beta=0.3, i=1$]14.542,20.797[	200	2000
$\beta=0.3, i=3$]21.788,25.115[	94	994
$\beta=0.3, i=4$]25.115,30.231[	106	1006
$\beta=0.5, i=1$]13.583,21.147[	200	2000
$\beta=0.5, i=4$]24.287,25.066[	88	988
$\beta=0.5, i=5$]25.066,30.344[	112	1012

This correction changes nothing about the band widths, the step-1 reconstruction errors, or the qualitative conclusions of the report (the success rate remains 100% in every non-empty interval for the original free-weight chaining): only the number of tested anchors had been misreported. The main Word document will be corrected accordingly.

1. New propagation rule

In the original algorithm (report, Section 7), each step of the chaining freely searches for the best pair (N_1, N_2) — both bounded by $N_{\text{MAX}} = 65536$ — to link the current anchor b_1 to a new partner b_2 reconstructing the next target. Only the value b_2 was reused as the anchor for the next step; the weight N_1 was re-optimized at every step, independently of what preceded it.

The requested variant modifies this rule: the weight assigned to the partner found at step n (i.e. $N_{2,n}$) becomes the FIXED weight of the anchor at step $n+1$ ($N_{1,\{n+1\}}$), at the same time as $b_{2,n}$ becomes $b_{1,\{n+1\}}$:

Step 1: free search $\rightarrow$ obtaining $(N_{1,1}, b_{1,1})$ and $(N_{2,1}, b_{2,1})$ [$b_{1,1}$ = starting anchor, method identical to the original report]

Step 2: $\{N_{1,2}, b_{1,2}\} \leftarrow \{N_{2,1}, b_{2,1}\}$, then search for $(N_{2,2}, b_{2,2})$ such that $(N_{1,2} \cdot b_{1,2} + N_{2,2} \cdot b_{2,2}) / (N_{1,2} + N_{2,2}) = x_{0,2}$, with $N_{1,2}$ now FIXED (= $N_{2,1}$)

Step 3: $\{N_{1,3}, b_{1,3}\} \leftarrow \{N_{2,2}, b_{2,2}\}$, then search for $(N_{2,3}, b_{2,3})$ for $x_{0,3}$, with $N_{1,3}$ FIXED (= $N_{2,2}$)

... and so on to the end of the list: at every step $n \geq 2$, $N_{1,n} = N_{2,\{n-1\}}$.

From step 2 onward, only a single degree of freedom remains (N_2 , a positive integer bounded by 65536) instead of two: for each candidate b_2 in the pool (opposite sign to b_1), we compute $N_{2,\text{exact}} = N_1 \cdot (x_0 - b_1) / (b_2 - x_0)$, round it to the nearest integer (bounded to $[1, 65536]$), and keep the candidate b_2 that minimizes the reconstruction error. This search is exhaustive over the entire pool at every step (no shortcut via continued-fraction approximation is possible, since b_2 is no longer a free variable).

Step 1 of every chain is taken unchanged from the original files (chainage_beta03.txt / chainage_beta05.txt), in order to guarantee exact continuity with the previous report; only step 2 onward has been recomputed with the new rule, on the extended pool ($t_{\max}=1000$, $k_{\text{range}}=15$, 60000 candidates), regenerated and independently verified (control value $t=9$, $k=-3$, $\sigma=-1 \rightarrow b=25.092889841584267$, identical to the report).

2. Main result: precision degrades, but remains very good overall

The computation was carried out exhaustively over the 8000 starting chains of the 6 non-empty founding intervals (2000+994+1006 for $\beta=0.3$; 2000+988+1012 for $\beta=0.5$), i.e. approximately 44,000 search steps, each scanning the entire pool of opposite sign ($\sim 30,000$ candidates).

Result: removing one degree of freedom (N_1 fixed and inherited rather than freely searched) markedly degrades the typical precision (median around 10^{-8} instead of 10^{-10} to 10^{-13} in the free case), and above all reveals, in certain intervals, a non-negligible fraction of chains whose error at a given step greatly exceeds 10^{-6} (up to a few % to nearly 0.2–0.3 in the isolated worst case). The proportion of chains that stay under 10^{-6} at EVERY step varies strongly by founding interval — from 40.7% to 99.8% — whereas it was 100% everywhere in the free case:

List	Founding interval	# chains	All steps < 1e-6	All steps < 1e-3
$\beta=0.3$	$i=1$]14.542,20.797[	2000	1950 (97.50%)	2000 (100.00%)
$\beta=0.3$	$i=3$]21.788,25.115[	994	692 (69.62%)	993 (99.90%)
$\beta=0.3$	$i=4$]25.115,30.231[	1006	1003 (99.70%)	1006 (100.00%)
$\beta=0.5$	$i=1$]13.583,21.147[	2000	1786 (89.30%)	1996 (99.80%)
$\beta=0.5$	$i=4$]24.287,25.066[	988	402 (40.69%)	985 (99.70%)
$\beta=0.5$	$i=5$]25.066,30.344[	1012	1010 (99.80%)	1011 (99.90%)

On the other hand, with a looser tolerance threshold (error $< 10^{-3}$), the success rate remains very high everywhere ($\geq 99.7\%$). The constraint introduced by weight inheritance therefore genuinely degrades the achievable precision, but does not eliminate the ability to chain: it simply makes it less consistently "perfect".

3. Why the degradation stays moderate: the role of the inherited weight N_1

The explanation lies in the order of magnitude of the inherited weight N_1 across steps. Unlike N_{1_1} (always large, since it comes from an optimized free search), the N_1 inherited at later steps is simply the N_2 found at the previous step — a value that is not optimized on any "stay large" criterion: it can occasionally fall very low. On the founding interval $\beta=0.3$ [1], the distribution of the inherited N_1 at each step is:

Step	Inherited N_1 — min	Inherited N_1 — p25	Inherited N_1 — median	Inherited N_1 — p75	Inherited N_1 — max
1	2035	32949	46134	56718	65531
2	289	4688	6782	27056	39169
3	12	5122	11146	16507	50315
4	42	2203	6628	26391	65529
5	50	6521	20888	39438	65527
6	124	4752	14492	25383	56756

The observed minimum drops quickly (from 2035 at step 1 to only 12 at step 3), while the median stays on the order of several thousand to a few tens of thousands. It is this dispersion — not a simple systematic shift — that explains the table in Section 2: most chains inherit an N_1 still large enough that the pool of $\sim 30,000$ candidates offers, by sheer combinatorial density, a point falling very close to the target; but a minority of chains inherit a

small N_1 (a few dozen to a few hundred), for which the effective resolution of the search collapses and the error explodes (up to 10^{-1} or more in the worst cases). A moderate negative correlation is measured between $\log(N_1)$ and $\log(\text{error})$ (coefficient ≈ -0.25), confirming this mechanism.

This result directly extends the critique in Section 6 of the main report, but in a more nuanced way than expected: removing one degree of freedom does introduce a real, measurable constraint (it produces occasional failures, absent from the free case), but it is not sufficient to systematically turn the problem into a severe test, because the inherited weight N_1 is not constrained to stay small at any step — it simply fluctuates, occasionally taking low values. A more severe and systematic test would consist of explicitly capping N_{MAX} to a small, fixed value from step 1 onward (for example a few hundred), so that the constraint applies uniformly to all chains rather than depending on a random inherited draw.

4. Detail by step and by founding interval

4.1 List $\beta=0.3$

Interval [1]]14.542, 20.797[— 2000 chains, 6 remaining steps:

Step	Target x_0	Min error	Median error	Mean error	Max error	% < $1e-6$
1	20.797	2.49e-14	1.13e-10	1.53e-10	1.06e-09	100.0%
2	21.788	3.68e-11	2.58e-08	4.98e-08	1.84e-06	99.75%
3	25.115	2.03e-11	4.13e-08	1.20e-07	3.33e-06	98.05%
4	30.231	8.40e-12	1.32e-08	3.67e-08	1.35e-06	99.95%
5	30.884	8.99e-13	1.76e-08	3.59e-08	1.45e-06	99.95%
6	33.514	4.46e-11	3.78e-08	8.10e-08	6.63e-06	99.80%

Interval [3]]21.788, 25.115[— 994 chains, 4 remaining steps:

Step	Target x_0	Min error	Median error	Mean error	Max error	% < $1e-6$
1	25.115	5.68e-14	1.07e-10	1.38e-10	1.12e-09	100.0%
2	30.231	2.86e-10	3.86e-07	1.39e-03	1.38e+00	79.07%
3	30.884	1.86e-11	3.47e-07	1.44e-06	8.42e-04	85.01%
4	33.514	1.59e-11	2.03e-08	8.00e-08	3.94e-05	99.80%

Interval [4]]25.115, 30.231[— 1006 chains, 3 remaining steps:

Step	Target x_0	Min error	Median error	Mean error	Max error	% < $1e-6$
1	30.231	3.20e-13	8.86e-11	1.15e-10	7.09e-10	100.0%
2	30.884	3.83e-12	1.52e-08	2.59e-08	5.37e-07	100.0%
3	33.514	1.93e-11	3.10e-08	6.40e-08	1.67e-06	99.70%

4.2 List $\beta=0.5$

Interval [1]]13.583, 21.147[— 2000 chains, 8 remaining steps:

Step	Target x_0	Min error	Median error	Mean error	Max error	% < $1e-6$
1	21.147	3.55e-14	9.95e-11	1.41e-10	1.10e-09	100.0%
2	21.760	4.47e-12	1.58e-08	3.06e-08	1.19e-06	99.95%
3	24.287	5.61e-12	2.53e-08	4.53e-08	2.86e-06	99.85%

Step	Target x0	Min error	Median error	Mean error	Max error	% < 1e-6
4	25.066	1.39e-11	3.41e-08	6.34e-08	1.91e-06	99.75%
5	30.344	3.95e-11	3.18e-08	3.32e-06	3.60e-03	94.20%
6	30.895	3.88e-12	2.84e-08	1.21e-06	8.22e-04	94.60%
7	32.787	5.17e-12	4.08e-08	7.18e-07	2.81e-04	95.20%
8	33.596	4.14e-12	2.91e-08	6.92e-05	1.05e-01	93.35%

Interval [4]]24.287, 25.066[— 988 chains, 5 remaining steps:

Step	Target x0	Min error	Median error	Mean error	Max error	% < 1e-6
1	25.066	2.84e-14	1.03e-10	1.38e-10	8.32e-10	100.0%
2	30.344	5.86e-10	5.76e-07	4.41e-05	2.80e-02	67.81%
3	30.895	1.12e-09	4.54e-07	3.00e-06	2.02e-03	77.53%
4	32.787	2.21e-10	1.44e-07	3.74e-07	8.36e-05	94.53%
5	33.596	7.15e-10	4.71e-07	3.72e-05	3.16e-02	69.64%

Interval [5]]25.066, 30.344[— 1012 chains, 4 remaining steps:

Step	Target x0	Min error	Median error	Mean error	Max error	% < 1e-6
1	30.344	1.78e-14	7.52e-11	5.55e-10	4.56e-07	100.0%
2	30.895	9.77e-13	1.32e-08	1.14e-06	1.13e-03	99.80%
3	32.787	1.62e-11	1.51e-08	3.31e-07	3.07e-04	99.90%
4	33.596	8.54e-11	2.31e-08	2.13e-04	2.16e-01	99.90%

Column note: "% < 1e-6" denotes the proportion of all chains in the interval whose reconstruction error at THIS step (individually) remains below 10^{-6} — including chains that already failed at an earlier step, since the chaining always continues with the best candidate found, even when that candidate is of mediocre precision. The degradation is particularly marked on intervals $\beta=0.3$ [3] and $\beta=0.5$ [4] (79% and 68% at the first post-inheritance step), whereas intervals [4]/[1]/[5] stay very close to 100% at every step — a heterogeneity that simply reflects the variability of the inherited N1 described in Section 3, specific to each population of chains.

5. Detailed examples (starting chain $l=1$, same starting point for both lists)

To concretely illustrate the mechanism, here is the full detail of the first starting chain ($l=1$) for each list, starting from the same initial anchor $b(t=1, k=-2, \sigma=+1) = 19.905401285$ (founding interval common to both lists). The "N1 (inherited)" column explicitly shows the requested propagation: at every line $n \geq 2$, the value of N1 is exactly equal to the N2 found on line $n-1$.

5.1 List $\beta=0.3$ — interval [1]

Step	Target x0	N1 (inherited)	N2 (found)	(t,k, σ)	b	Error
0 (start)	—	—	—	$t=1, k=-2, +$	19.905401285	—
1	20.797	21352	4591	$t=995, k=-3, -$	24.943681718	2.61e-11
2	21.788	4591	10197	$t=304, k=-2, +$	20.367215913	3.84e-08

Step	Target x0	N1 (inherited)	N2 (found)	(t,k,σ)	b	Error
3	25.115	10197	2694	t=306,k=-5,-	43.085733100	4.95e-08
4	30.231	2694	43861	t=687,k=-3,+	29.441445489	1.25e-08
5	30.884	43861	20157	t=268,k=-4,-	34.022953343	1.34e-08
6	33.514	20157	2519	t=681,k=-3,+	29.441362434	6.58e-08

Line-by-line check: N2 of step 1 = 4591 → N1 of step 2 = 4591; N2 of step 2 = 10197 → N1 of step 3 = 10197; and so on through step 6. This chain stays very precise at every step (maximum error 6.6×10^{-8}), the inherited N1 never dropping below ~2700.

5.2 List $\beta=0.5$ — interval [1]

Step	Target x0	N1 (inherited)	N2 (found)	(t,k,σ)	b	Error
0 (start)	—	—	—	t=1,k=-2,+	19.905401285	—
1	21.147	32362	10583	t=991,k=-3,-	24.943713372	1.68e-11
2	21.760	10583	24097	t=214,k=-2,+	20.361766279	1.87e-08
3	24.287	24097	2562	t=673,k=-7,-	61.205952865	1.02e-08
4	25.066	2562	19635	t=120,k=-2,+	20.350412644	6.77e-08
5	30.344	19635	53455	t=501,k=-4,-	34.014827631	4.47e-08
6	30.895	53455	2254	t=187,k=5,+	-43.093637124	3.73e-08
7	32.787	2254	16611	t=361,k=-5,-	43.083487444	1.67e-07
8	33.596	16611	3185	t=524,k=2,+	-15.884896031	3.56e-09

Same check: N2 of step 1 = 10583 → N1 of step 2 = 10583; N2 of step 2 = 24097 → N1 of step 3 = 24097; etc. This particular chain also stays very precise over its 8 steps (maximum error 1.7×10^{-7}).

6. Conclusion of this addendum

The propagation rule tested here — inheriting at every step the weight N2 found at the previous step as the new fixed N1 — introduces a real, measurable constraint: the typical precision degrades by 2 to 4 orders of magnitude compared with the fully free case, and above all, a significant fraction of chains (up to about 60% on the most affected interval, $\beta=0.5$ [4]) fail to stay under the 10^{-6} threshold at every step — a phenomenon completely absent from the free-weight chaining (100% success everywhere). With a more realistic tolerance threshold (10^{-3}), the overall success rate nonetheless remains very high ($\geq 99.7\%$ everywhere).

The identified cause is that the inherited weight N1 fluctuates strongly from one chain to another (from a few dozen to several tens of thousands), without ever being systematically constrained to stay small: when it remains large (the majority case), the combinatorial-density argument from Section 6 of the main report continues to apply almost intact; when it drops low (a minority but non-negligible case), the precision collapses locally. To turn this variant into a truly discriminating test of the DMT framework, one would need to combine weight inheritance with a much smaller, fixed N_{MAX} from step 1 onward (for example a few hundred), so that the constraint applies uniformly to all chains rather than depending on a randomly inherited draw whose magnitude varies from chain to chain.